

PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 8

APPLICATIONS OF VECTORS IN MECHANICS

SUGGESTED TIME: 11 PERIODS

IN THIS CHAPTER

This chapter extends differentiation and integration from scalars to vectors and applies them to the kinematics of a particle: displacement, velocity and acceleration as vector quantities. It provides the essential mathematical machinery for the whole of GCE Advanced Level mechanics.

- By the end of this chapter I can differentiate a vector given in component form with respect to a scalar parameter (usually time).
- By the end of this chapter I can interpret the derivative of a position vector physically as velocity, and the derivative of velocity as acceleration.
- By the end of this chapter I can integrate a vector function component-wise, including definite integrals, and determine constants of integration from initial conditions.
- By the end of this chapter I can obtain velocity and displacement vectors from a given acceleration vector by integration.
- By the end of this chapter I can obtain velocity and acceleration vectors from a given displacement vector by differentiation.
- By the end of this chapter I can use the chain rule on vectors, e.g. differentiate $\mathbf{r}(t)$ where components are composite functions of t .



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1. Differentiating vectors: the mechanics of vector calculus
2. Physical applications of vector differentiation: velocity and acceleration
3. Integrating vectors: indefinite and definite integrals
4. Displacement, velocity and acceleration: the kinematic triangle of calculus
5. From differentiation to velocity and acceleration vectors (Lesson 88)
6. From integration to velocity and displacement vectors (Lesson 91) and synthesis
7. Integration activity
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CHAPTER OBJECTIVES

- By the end of this chapter I can differentiate a vector given in component form with respect to a scalar parameter (usually time).
- By the end of this chapter I can interpret the derivative of a position vector physically as velocity, and the derivative of velocity as acceleration.
- By the end of this chapter I can integrate a vector function component-wise, including definite integrals, and determine constants of integration from initial conditions.
- By the end of this chapter I can obtain velocity and displacement vectors from a given acceleration vector by integration.
- By the end of this chapter I can obtain velocity and acceleration vectors from a given displacement vector by differentiation.
- By the end of this chapter I can use the chain rule on vectors, e.g. differentiate $\mathbf{r}(t)$ where components are composite functions of t .
- By the end of this chapter I can solve kinematics problems in two dimensions using $\mathbf{i}, \mathbf{j}$ components (projectile-type and general plane motion).
- By the end of this chapter I can distinguish clearly between displacement and distance, and between velocity and speed, in vector contexts.
- By the end of this chapter I can present a complete, well-structured GCE-style solution linking $\mathbf{r}$, $\mathbf{v}$ and $\mathbf{a}$ through calculus.

PREREQUISITES

- Scalar differentiation and integration from Lower Sixth pure mathematics (polynomials, trigonometric and exponential functions, chain rule).
- Vector algebra from Lower Sixth: components in i , j (and k), magnitude of a vector, unit vectors, scalar product.
- Basic kinematics from Lower Sixth mechanics: constant acceleration formulae (su-vat) in one dimension.
- Trigonometry: resolving quantities into perpendicular components.
- Sketching and interpreting graphs of motion (displacement–time, velocity–time).

1 DIFFERENTIATING VECTORS: THE MECHANICS OF VECTOR CALCULUS

A bush taxi leaves the Bafoussam motor park and the driver accelerates, brakes, and negotiates bends on the road to Bamenda. At every instant, the vehicle has a position, a velocity and an acceleration — and none of these is just a number: each has a **direction** in space. If the driver only knows his speed from the speedometer, can he predict where the taxi will be ten seconds later around a curve? Clearly not: two vehicles with the same speed of 60 km h^{-1} but different directions will be in very different places after ten seconds. To answer such questions, physicists **differentiate** and **integrate vectors**, exactly as we differentiate and integrate scalars in pure mathematics. This chapter gives you the calculus of vectors that powers all of mechanics.

1.1 Vector-valued functions: position as a function of time

Imagine the bush taxi moving in a horizontal plane. Choose an origin O (say, the motor park gate) and fixed perpendicular directions $\vec{i}$ (east) and $\vec{j}$ (north). At time t , the position of the taxi is described by the **position vector**

$$\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j},$$

where $x(t)$ and $y(t)$ are ordinary scalar functions of t . (The same ideas extend directly to three dimensions with an additional $\vec{k}$ component.)

DEFINITION Vector-valued function

A **vector-valued function** of a scalar parameter t is a rule that assigns to each value of t (in some interval) a vector

$$\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}.$$

The scalar functions $x(t)$ and $y(t)$ are called the **components** (or coordinates) of $\vec{r}$. As t varies, the tip of $\vec{r}(t)$ traces a curve in the plane called the **path** or **trajectory**.

In mechanics the parameter t is almost always **time**, measured in seconds, and the components are measured in metres. But the mathematics works for any parameter — an angle, a distance, etc.

EXAMPLE A first vector function

Let $\vec{r}(t) = 3t\vec{i} + (5 - t^2)\vec{j}$. At $t = 0$, $\vec{r}(0) = 5\vec{j}$; at $t = 1$, $\vec{r}(1) = 3\vec{i} + 4\vec{j}$; at $t = 2$, $\vec{r}(2) = 6\vec{i} + \vec{j}$. Plotting these tips gives the beginning of a curve — the path of the moving point.

1.2 The derivative of a vector function

Intuition. Suppose the tip of $\vec{r}(t)$ is at point P at time t , and at point Q at the later time $t + \Delta t$. The

change in position is

$$\Delta \vec{r} = \vec{r}(t + \Delta t) - \vec{r}(t) = \vec{PQ},$$

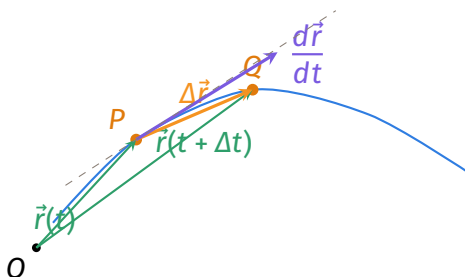
the chord from P to Q . The quotient $\frac{\Delta \vec{r}}{\Delta t}$ is the average rate of change of position over the interval. As $\Delta t \rightarrow 0$, the point Q slides along the curve towards P , the chord $\vec{PQ}$ swings round towards the **tangent** at P , and the quotient settles down to a limiting vector: the derivative.

DEFINITION Derivative of a vector function

The derivative of $\vec{r}(t)$ with respect to t is the limit

$$\frac{d\vec{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t},$$

whenever this limit exists. It is itself a vector, and it points **tangent to the path** at the point $\vec{r}(t)$, in the direction of increasing t . The notation $\dot{\vec{r}}$ is also used for $\frac{d\vec{r}}{dt}$.



As $\Delta t \rightarrow 0$, the chord $\Delta \vec{r} = \vec{PQ}$ shrinks and turns towards the tangent: the derivative $\frac{d\vec{r}}{dt}$ is tangent to the path at P .

The practical rule. Because $\vec{i}$ and $\vec{j}$ are fixed vectors, they pass through the limit unchanged. Writing out the components,

$$\frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t} = \frac{x(t + \Delta t) - x(t)}{\Delta t} \vec{i} + \frac{y(t + \Delta t) - y(t)}{\Delta t} \vec{j}.$$

Letting $\Delta t \rightarrow 0$, each scalar quotient tends to the ordinary derivative of that component. Hence the fundamental working rule:

PROPERTY Differentiate component by component

If $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$, then

$$\frac{d\vec{r}}{dt} = \frac{dx}{dt} \vec{i} + \frac{dy}{dt} \vec{j}.$$

That is: differentiate each component separately and keep $\vec{i}$ and $\vec{j}$ fixed. The derivation

from the limit definition (sketched above) is instructive and may be asked as a short proof.

METHOD How to differentiate a vector

1. Identify the parameter (usually time t) and write $\vec{r}(t)$ in component form.
2. Differentiate the x -component with respect to t : this gives the $\vec{i}$ -component of $\frac{d\vec{r}}{dt}$.
3. Differentiate the y -component with respect to t : this gives the $\vec{j}$ -component.
4. Leave the unit vectors $\vec{i}, \vec{j}$ untouched — they are constant.

WARNING Do not differentiate the unit vectors

A very common mistake is to apply the product rule to $x(t)\vec{i}$ and write something like $\frac{dx}{dt}\vec{i} + x\frac{d\vec{i}}{dt}$ with an unknown $\frac{d\vec{i}}{dt}$. In this course $\vec{i}$ and $\vec{j}$ are **fixed** in space, so

$$\frac{d\vec{i}}{dt} = \vec{0} \quad \text{and} \quad \frac{d\vec{j}}{dt} = \vec{0}.$$

(Rotating unit vectors have non-zero derivatives, but that belongs to more advanced mechanics and is not in the syllabus.)

1.3 Rules of vector differentiation

All the familiar rules of scalar calculus survive, because differentiation acts on each component separately.

PROPERTY Linearity

For any vector functions $\vec{a}(t), \vec{b}(t)$ and constants $\alpha, \beta \in \mathbb{R}$,

$$\frac{d}{dt}(\alpha \vec{a} + \beta \vec{b}) = \alpha \frac{d\vec{a}}{dt} + \beta \frac{d\vec{b}}{dt}.$$

In words, **differentiation of vectors is linear**: the derivative of a sum is the sum of the derivatives, and constant factors pass through.

PROPERTY Product rule for a scalar times a vector

If $f(t)$ is a scalar function and $\vec{r}(t)$ a vector function, then

$$\frac{d}{dt}[f(t)\vec{r}(t)] = \frac{df}{dt}\vec{r}(t) + f(t)\frac{d\vec{r}}{dt}.$$

This follows by applying the ordinary product rule to each component of $f\vec{r} = (fx)\vec{i} + (fy)\vec{j}$.

PROPERTY Chain rule

If $\vec{r}$ depends on a variable u , and u itself depends on t (i.e. $u = u(t)$), then

$$\frac{d\vec{r}}{dt} = \frac{d\vec{r}}{du} \frac{du}{dt}.$$

The scalar factor $\frac{du}{dt}$ simply multiplies the vector $\frac{d\vec{r}}{du}$.

EXAMPLE Using the product and chain rules

Let $\vec{r}(t) = t^2(3t\vec{i} + 2\vec{j})$.

- Product rule: with $f(t) = t^2$ and $\vec{a}(t) = 3t\vec{i} + 2\vec{j}$,

$$\frac{d\vec{r}}{dt} = 2t(3t\vec{i} + 2\vec{j}) + t^2(3\vec{i}) = 9t^2\vec{i} + 4t\vec{j}.$$

- Check by expanding first: $\vec{r}(t) = 3t^3\vec{i} + 2t^2\vec{j}$, so $\frac{d\vec{r}}{dt} = 9t^2\vec{i} + 4t\vec{j}$. ✓

1.4 Trigonometric and exponential components; higher derivatives

Circular and oscillatory motions produce trigonometric components; growth and decay produce exponentials. Nothing new is needed — differentiate each component with the rules of pure mathematics.

EXAMPLE Differentiating a circular motion

Let $\vec{r}(t) = \cos t\vec{i} + \sin t\vec{j}$ (the tip moves on the unit circle). Then

$$\frac{d\vec{r}}{dt} = -\sin t\vec{i} + \cos t\vec{j}.$$

Notice that $\vec{r} \cdot \frac{d\vec{r}}{dt} = -\cos t \sin t + \sin t \cos t = 0$: the derivative is perpendicular to the posi-

tion vector, exactly as a tangent to a circle is perpendicular to the radius. Differentiating again,

$$\frac{d^2\vec{r}}{dt^2} = -\cos t \vec{i} - \sin t \vec{j} = -\vec{r}(t),$$

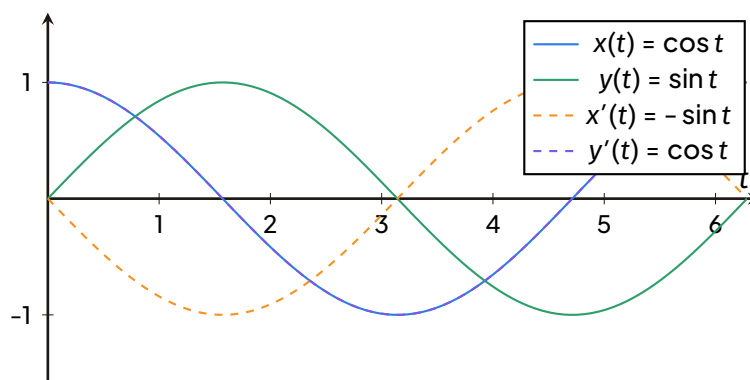
so the second derivative points back towards the origin — the first hint of centripetal acceleration, which we shall meet later.

DEFINITION Second derivative

The **second derivative** of $\vec{r}(t)$ is obtained by differentiating each component twice:

$$\frac{d^2\vec{r}}{dt^2} = \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right) = \frac{d^2x}{dt^2}\vec{i} + \frac{d^2y}{dt^2}\vec{j}.$$

The notation $\ddot{\vec{r}}$ is also used. Higher derivatives are defined similarly.



For $\vec{r}(t) = \cos t \vec{i} + \sin t \vec{j}$, each component (solid) is differentiated separately to give the components of $\frac{d\vec{r}}{dt}$ (dashed).

WORKED EXAMPLE 1 Computing $\frac{d\vec{r}}{dt}$ and $\frac{d^2\vec{r}}{dt^2}$ at a given time

A particle moves so that its position vector at time t seconds is

$$\vec{r}(t) = (2t^3 - 3t)\vec{i} + (4e^{-t} + t)\vec{j} \quad (\text{metres}).$$

- (a) Find $\frac{d\vec{r}}{dt}$ and $\frac{d^2\vec{r}}{dt^2}$. (b) Evaluate $\frac{d\vec{r}}{dt}$ at $t = 1$. (c) Find the value of $t > 0$ at which the $\vec{i}$ -component of $\frac{d\vec{r}}{dt}$ vanishes.

SOLUTION

(a) Differentiate each component, keeping $\vec{i}$ and $\vec{j}$ fixed:

$$\frac{d\vec{r}}{dt} = (6t^2 - 3)\vec{i} + (-4e^{-t} + 1)\vec{j}.$$

Differentiating again:

$$\frac{d^2\vec{r}}{dt^2} = 12t\vec{i} + 4e^{-t}\vec{j}.$$

(b) At $t = 1$:

$$\left. \frac{d\vec{r}}{dt} \right|_{t=1} = (6 - 3)\vec{i} + \left(1 - \frac{4}{e}\right)\vec{j} = 3\vec{i} + \left(1 - \frac{4}{e}\right)\vec{j} \approx 3\vec{i} - 0.47\vec{j}.$$

(c) The $\vec{i}$ -component of $\frac{d\vec{r}}{dt}$ is $6t^2 - 3$. It vanishes when

$$6t^2 - 3 = 0 \implies t^2 = \frac{1}{2} \implies t = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.707 \text{ s},$$

taking the positive root since $t > 0$.

WARNING GCE exam tips

- State the parameter. Always write explicitly that you differentiate with respect to time t . Marks are awarded for correct notation such as $\frac{d\vec{r}}{dt}$, $\dot{\vec{r}}$.
- Keep the vector form. Do not drop $\vec{i}$ or $\vec{j}$ in intermediate lines; an answer like " $6t^2 - 3 + (-4e^{-t} + 1)$ " confuses components and scores no marks.
- Check units. If $\vec{r}$ is in metres and t in seconds, then $\frac{d\vec{r}}{dt}$ is in m s^{-1} and $\frac{d^2\vec{r}}{dt^2}$ in m s^{-2} .
- Always write the unit vectors in your final answer for the derivative.

EXERCISE 1 Practice

1. Given $\vec{r}(t) = (3t^2 - 2t)\vec{i} + (t^3 + 5)\vec{j}$, find $\frac{d\vec{r}}{dt}$ and $\frac{d^2\vec{r}}{dt^2}$, and evaluate both at $t = 2$.
2. A point moves with $\vec{r}(t) = 4 \cos(2t)\vec{i} + 4 \sin(2t)\vec{j}$. Show that $\frac{d^2\vec{r}}{dt^2} = -4\vec{r}(t)$, and state what this means geometrically.
3. Given $\vec{r}(t) = e^{2t}\vec{i} + \ln(1+t)\vec{j}$, find the time t at which the $\vec{j}$ -component of $\frac{d\vec{r}}{dt}$ equals $\frac{1}{2}$.

SOLUTION Exercise 1

1. $\frac{d\vec{r}}{dt} = (6t - 2)\vec{i} + 3t^2\vec{j}$; $\frac{d^2\vec{r}}{dt^2} = 6\vec{i} + 6t\vec{j}$. At $t = 2$: $\frac{d\vec{r}}{dt} = 10\vec{i} + 12\vec{j}$ and $\frac{d^2\vec{r}}{dt^2} = 6\vec{i} + 12\vec{j}$.
2. $\frac{d\vec{r}}{dt} = -8\sin(2t)\vec{i} + 8\cos(2t)\vec{j}$; $\frac{d^2\vec{r}}{dt^2} = -16\cos(2t)\vec{i} - 16\sin(2t)\vec{j} = -4(4\cos(2t)\vec{i} + 4\sin(2t)\vec{j}) = -4\vec{r}(t)$. Geometrically, the second derivative always points from the particle towards the origin (opposite to $\vec{r}$).
3. $\frac{d\vec{r}}{dt} = 2e^{2t}\vec{i} + \frac{1}{1+t}\vec{j}$. Setting $\frac{1}{1+t} = \frac{1}{2}$ gives $1+t = 2$, so $t = 1$.

KEY POINTS Key points of this section

- A vector-valued function $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$ assigns a vector to each value of the parameter t ; its tip traces the path of the moving point.
- The derivative is defined by $\frac{d\vec{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t}$ and is computed component by component: $\frac{d\vec{r}}{dt} = \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j}$.
- Geometrically, $\frac{d\vec{r}}{dt}$ is **tangent** to the path, pointing in the direction of motion.
- Differentiation is linear; the product rule $\frac{d}{dt}[f\vec{r}] = f'\vec{r} + f\vec{r}'$ and the chain rule $\frac{d\vec{r}}{dt} = \frac{d\vec{r}}{du} \cdot \frac{du}{dt}$ hold exactly as for scalars.
- The second derivative $\frac{d^2\vec{r}}{dt^2}$ is obtained by differentiating each component twice.
- The unit vectors $\vec{i}$ and $\vec{j}$ are fixed: their derivatives are zero.

DID YOU KNOW? Why this matters for the road to Bamenda

Every navigation system — from the GPS in a Douala taxi to the tracking of an aircraft over Mount Cameroon — performs exactly the calculations of this section. The position is stored as a vector function of time; differentiating gives velocity, differentiating again gives acceleration. In the next section we shall give these derivatives their physical names and see Newton's laws come alive through vector calculus.

2 PHYSICAL APPLICATIONS OF VECTOR DIFFERENTIATION: VELOCITY AND ACCELERATION

In the previous section we mastered the mechanics of differentiating a vector function: because the basis vectors $\vec{i}$ and $\vec{j}$ are fixed in space, the derivative of $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$ is obtained by differentiating each component separately. We now give this operation its physical meaning. This is where vector calculus becomes kinematics — the mathematical description of motion. Almost every mechanics question in Paper 3 of the GCE Advanced Level that involves a moving particle relies on the ideas developed in this section.

2.1 The position vector and displacement

Consider a particle P moving in a plane. At each instant t the particle occupies a point of the plane, and we describe that point by its position relative to a fixed origin O .

DEFINITION Position vector

The **position vector** of a particle P at time t , relative to a fixed origin O , is

$$\vec{r}(t) = \vec{OP} = x(t)\vec{i} + y(t)\vec{j},$$

where $x(t)$ and $y(t)$ are the Cartesian coordinates of P at time t . As t varies, the tip of $\vec{r}(t)$ traces out the **path** (or trajectory) of the particle.

DEFINITION Displacement

The **displacement** of the particle between times t_1 and t_2 is the vector

$$\Delta\vec{r} = \vec{r}(t_2) - \vec{r}(t_1).$$

It is a vector joining the initial position to the final position, independent of the actual path taken.

Think of a pirogue crossing the Wouri estuary: at each moment its position can be given by two numbers (distances east and north of the jetty), and as time passes these two numbers change, drawing the curve of the journey. The displacement from departure to arrival is simply the straight-line vector from the jetty to the landing point.

2.2 Velocity: the rate of change of displacement

Intuition. Suppose that between times t and $t + \delta t$ the particle moves from P to Q . Its displacement during this short interval is the vector

$$\delta\vec{r} = \vec{r}(t + \delta t) - \vec{r}(t) = \vec{PQ}.$$

Dividing by the elapsed time δt gives the average velocity over the interval, a vector parallel to

the chord PQ . As $\delta t \rightarrow 0$, the point Q slides along the path towards P , the chord PQ tends to the tangent to the path at P , and the average velocity tends to the instantaneous velocity. This limiting picture is the whole idea.

DEFINITION Velocity

The **velocity** of a particle at time t is the derivative of its position vector with respect to time:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = \dot{x}(t)\vec{i} + \dot{y}(t)\vec{j}.$$

The velocity vector is always tangent to the path at the position of the particle, and points in the direction of motion.

PROPERTY Speed

The **speed** of the particle is the magnitude of its velocity:

$$|\vec{v}(t)| = \sqrt{\dot{x}(t)^2 + \dot{y}(t)^2}.$$

Velocity is a vector (it has magnitude and direction); speed is a scalar (a non-negative number). Two particles can have the same speed while moving in completely different directions.

WARNING Exam vocabulary — GCE

- “Find the velocity” → give the vector $\vec{v} = \dot{x}\vec{i} + \dot{y}\vec{j}$.
- “Find the speed” → give the scalar $|\vec{v}| = \sqrt{\dot{x}^2 + \dot{y}^2}$.
- “Find the magnitude of the velocity” → same as speed.

Writing down only $\dot{x}$ or only $\dot{y}$ when speed is asked loses the mark.

2.3 Acceleration: the rate of change of velocity

Intuition. A car travelling along the winding road from Buea to Limbe may keep its speedometer fixed at 50 km h^{-1} , yet its velocity is changing continuously because its direction is changing. Any change of velocity — in magnitude or in direction — constitutes an acceleration.

DEFINITION Acceleration

The **acceleration** of a particle at time t is the derivative of its velocity, i.e. the second

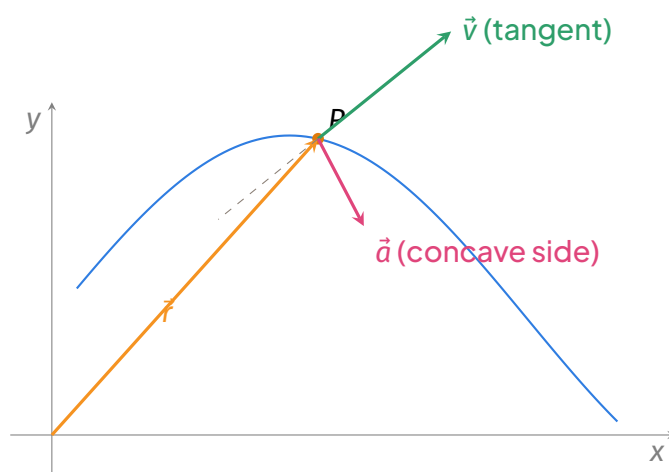
derivative of its position vector:

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = \ddot{x}(t)\vec{i} + \ddot{y}(t)\vec{j}.$$

A crucial geometric fact follows. Velocity is always tangent to the path, but acceleration is not: acceleration measures how the velocity vector itself is turning, and it points towards the **concave (inner) side** of the trajectory. Only when the path is a straight line do $\vec{v}$ and $\vec{a}$ share the same line of action.

WARNING Velocity direction $\neq$ acceleration direction

Do not confuse the direction of motion (the direction of $\vec{v}$, tangent to the path) with the direction of acceleration. They coincide only for straight-line motion. On a curved path, $\vec{a}$ always points towards the inside (concave side) of the curve, even when the particle is slowing down.



A particle path with the position vector $\vec{r}$, the velocity $\vec{v}$ tangent to the path, and the acceleration $\vec{a}$ pointing towards the concave side of the trajectory.

2.4 Finding velocity and acceleration from a position vector

The procedure is purely mechanical: differentiate each component once for $\vec{v}$, twice for $\vec{a}$, then substitute the required value of t .

METHOD From $\vec{r}(t)$ to $\vec{v}(t)$ and $\vec{a}(t)$

1. Write $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$.
2. Differentiate component-wise: $\vec{v}(t) = \dot{x}(t)\vec{i} + \dot{y}(t)\vec{j}$.
3. Differentiate again: $\vec{a}(t) = \ddot{x}(t)\vec{i} + \ddot{y}(t)\vec{j}$.
4. If a specific time t_0 is given, substitute t_0 into $\vec{v}$ and $\vec{a}$.
5. For speed or magnitude of acceleration, compute the square root of the sum of squares of the components.

WORKED EXAMPLE 2 Velocity, speed and acceleration from $\vec{r}(t)$

A particle moves so that its position vector at time t seconds is

$$\vec{r} = (2t^2 + 1)\vec{i} + (t^3 - 4t)\vec{j} \text{ metres.}$$

(a) Find the velocity and acceleration vectors at time t . (b) Find the velocity, the speed and the acceleration when $t = 2$.

SOLUTION

(a) Differentiate component by component:

$$\vec{v} = \frac{d\vec{r}}{dt} = 4t\vec{i} + (3t^2 - 4)\vec{j}, \quad \vec{a} = \frac{d\vec{v}}{dt} = 4\vec{i} + 6t\vec{j}.$$

(b) At $t = 2$:

$$\vec{v}(2) = 8\vec{i} + (12 - 4)\vec{j} = 8\vec{i} + 8\vec{j} \text{ m s}^{-1},$$

$$|\vec{v}(2)| = \sqrt{8^2 + 8^2} = \sqrt{128} = 8\sqrt{2} \approx 11.3 \text{ m s}^{-1},$$

$$\vec{a}(2) = 4\vec{i} + 12\vec{j} \text{ m s}^{-2}, \quad |\vec{a}(2)| = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10} \approx 12.6 \text{ m s}^{-2}.$$

Note that the speed is the magnitude of the whole velocity vector, not the value of one component.

2.5 Moving parallel to a given direction

A particle moves **parallel to the x -axis** exactly when its velocity has no $\vec{j}$ -component, i.e. when $\dot{y} = 0$ (with $\dot{x} \neq 0$). Similarly, motion parallel to the y -axis requires $\dot{x} = 0$ (with $\dot{y} \neq 0$). More generally, motion parallel to a direction $p\vec{i} + q\vec{j}$ requires the components of $\vec{v}$ to be in the ratio $\dot{x} : \dot{y} = p : q$, i.e. $q\dot{x} = p\dot{y}$.

METHOD When is the particle moving parallel to an axis or a given direction?

1. Differentiate $\vec{r}(t)$ to obtain $\vec{v} = \dot{x}\vec{i} + \dot{y}\vec{j}$.
2. For motion parallel to the x -axis, solve $\dot{y} = 0$; for the y -axis, solve $\dot{x} = 0$.
3. Check that the other component is non-zero at the solution (otherwise the particle is instantaneously at rest, not moving parallel to anything).
4. For a general direction $p\vec{i} + q\vec{j}$, solve $q\dot{x} = p\dot{y}$.

WORKED EXAMPLE 3 Moving parallel to the x -axis

A particle has position vector $\vec{r} = (t^2 - 3t)\vec{i} + (2t^3 - 6t)\vec{j}$ metres at time t s. Find the time(s) at which the particle is moving parallel to the x -axis, and state its position at that instant.

SOLUTION

Differentiate: $\vec{v} = (2t - 3)\vec{i} + (6t^2 - 6)\vec{j}$.

Motion parallel to the x -axis requires the $\vec{j}$ -component to vanish:

$$6t^2 - 6 = 0 \implies t^2 = 1 \implies t = 1 \quad (t \geq 0).$$

Check the $\vec{i}$ -component at $t = 1$: $2(1) - 3 = -1 \neq 0$, so the particle is genuinely moving (in the negative x -direction). Its position is

$$\vec{r}(1) = (1 - 3)\vec{i} + (2 - 6)\vec{j} = -2\vec{i} - 4\vec{j} \text{ metres.}$$

2.6 Enrichment: uniform circular motion — a preview

A beautiful application — which you will meet again in the chapter on circular motion — is the particle moving with constant angular speed ω on a circle of radius R centred at O :

$$\vec{r} = R \cos(\omega t)\vec{i} + R \sin(\omega t)\vec{j}.$$

Differentiating twice:

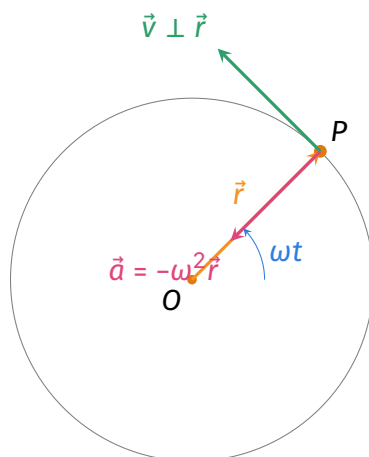
$$\vec{v} = -R\omega \sin(\omega t)\vec{i} + R\omega \cos(\omega t)\vec{j}, \quad \vec{a} = -R\omega^2 \cos(\omega t)\vec{i} - R\omega^2 \sin(\omega t)\vec{j} = -\omega^2 \vec{r}.$$

Two remarkable facts appear. First,

$$\vec{r} \cdot \vec{v} = -R^2\omega \cos(\omega t) \sin(\omega t) + R^2\omega \sin(\omega t) \cos(\omega t) = 0,$$

so the velocity is **perpendicular to the radius** (tangent to the circle), with constant speed $|\vec{v}| = R\omega$. Second, $\vec{a} = -\omega^2\vec{r}$: the acceleration is directed along the radius **towards the centre** — this is the famous **centripetal acceleration**, of magnitude $\omega^2 R = \frac{|\vec{v}|^2}{R}$. Here the speed is constant,

yet the acceleration is not zero: a perfect illustration that acceleration detects changes of direction as well as changes of speed.



Uniform circular motion: $\vec{v}$ is perpendicular to $\vec{r}$ (tangent to the circle) while $\vec{a}$ points towards the centre O .

DID YOU KNOW? Why this matters beyond the exam

The same computation governs a satellite in a circular orbit around the Earth: its velocity is tangent to the orbit while its acceleration — provided by gravity — points towards the centre of the Earth. The mathematics you have just used on $\vec{r} = R(\cos \omega t \vec{i} + \sin \omega t \vec{j})$ is literally the mathematics of space navigation.

KEY POINTS Key points

- Position: $\vec{r} = x\vec{i} + y\vec{j}$; velocity: $\vec{v} = \frac{d\vec{r}}{dt} = \dot{x}\vec{i} + \dot{y}\vec{j}$; acceleration: $\vec{a} = \frac{d\vec{v}}{dt} = \ddot{x}\vec{i} + \ddot{y}\vec{j}$.
- $\vec{v}$ is tangent to the path; speed = $|\vec{v}| = \sqrt{\dot{x}^2 + \dot{y}^2}$ is a scalar.
- $\vec{a}$ points towards the concave side of the trajectory; it coincides with the direction of motion only for straight-line paths.
- Motion parallel to the x -axis: solve $\dot{y} = 0$ (and check $\dot{x} \neq 0$); parallel to the y -axis: solve $\dot{x} = 0$ (and check $\dot{y} \neq 0$).
- Circular motion: $\vec{r} = R(\cos \omega t \vec{i} + \sin \omega t \vec{j})$ gives $\vec{v} \perp \vec{r}$ and $\vec{a} = -\omega^2 \vec{r}$ (centripetal).

EXERCISE 2 Practice

A particle moves with position vector $\vec{r} = (t^3 - 3t)\vec{i} + (t^2 + 2)\vec{j}$ metres at time t s.

1. Find $\vec{v}$ and $\vec{a}$ at time t .
2. Find the speed of the particle when $t = 2$.
3. Find the time at which the particle is moving parallel to the y -axis.
4. Show that the acceleration is not constant in direction, but that its $\vec{j}$ -component is constant. State the direction of the acceleration at $t = 0$.

SOLUTION Exercise 1

1. $\vec{v} = (3t^2 - 3)\vec{i} + 2t\vec{j}$; $\vec{a} = 6t\vec{i} + 2\vec{j}$.
2. At $t = 2$: $\vec{v} = 9\vec{i} + 4\vec{j}$, so $|\vec{v}| = \sqrt{81 + 16} = \sqrt{97} \approx 9.85 \text{ m s}^{-1}$.
3. Parallel to the y -axis: $\dot{x} = 0 \Rightarrow 3t^2 - 3 = 0 \Rightarrow t = 1$ (taking $t \geq 0$). Check: $\dot{y} = 2 \neq 0$, so the motion is genuinely vertical.
4. $\vec{a} = 6t\vec{i} + 2\vec{j}$. The direction of $\vec{a}$ is given by the ratio of its components: $\tan \theta = \frac{2}{6t} = \frac{1}{3t}$, which depends on t . Hence the direction is not constant. However, the $\vec{j}$ -component is the constant 2 m s^{-2} . At $t = 0$, $\vec{a} = 2\vec{j}$, so the acceleration is purely in the positive y -direction (parallel to the y -axis). As t increases, the direction rotates towards the positive x -axis.

3 INTEGRATING VECTORS: INDEFINITE AND DEFINITE INTEGRALS

In the previous section we saw that differentiation takes us down the kinematic chain: from the position vector $\vec{r}(t)$ we obtained the velocity $\vec{v}(t) = \dot{\vec{r}}(t)$, and from the velocity we obtained the acceleration $\vec{a}(t) = \dot{\vec{v}}(t)$. In mechanics, however, the information we actually measure is very often the acceleration (from the forces acting, through Newton's second law) or the velocity (from a speedometer, a radar gun, or a ticker-timer in the laboratory). The natural question is therefore the reverse one: given $\vec{a}(t)$, can we recover $\vec{v}(t)$? Given $\vec{v}(t)$, can we recover $\vec{r}(t)$?

Since differentiation of a vector is done component by component, it is entirely natural that integration — the inverse operation — should also be done component by component. This section makes that idea precise, shows how the constant of integration becomes a vector, and explains how initial conditions fix that constant. This is the mathematical engine behind Lessons 82, 83 and 91 of the syllabus.

3.1 Definition of the integral of a vector function

DEFINITION Indefinite integral of a vector function

Let $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$ be a vector function of the scalar variable t , whose components $x(t)$ and $y(t)$ are integrable. The **indefinite integral** of $\vec{r}$ with respect to t is obtained by integrating each component separately:

$$\int \vec{r}(t) dt = \left(\int x(t) dt \right) \vec{i} + \left(\int y(t) dt \right) \vec{j} + \vec{c},$$

where $\vec{c}$ is a constant vector of integration. In three dimensions, with $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$, we simply add the term $(\int z(t) dt)\vec{k}$.

The intuition is exactly the one you already have from scalar calculus: integration undoes differentiation. Because the unit vectors $\vec{i}, \vec{j}, \vec{k}$ are constant (they do not depend on t), they behave like ordinary constants during integration and can be taken outside the integral sign:

$$\int [x(t)\vec{i} + y(t)\vec{j}] dt = \vec{i} \int x(t) dt + \vec{j} \int y(t) dt.$$

PROPERTY The constant of integration is a vector

When a vector function is integrated indefinitely, the constant of integration is itself a **constant vector**:

$$\vec{c} = c_1 \vec{i} + c_2 \vec{j} \quad (\text{and } + c_3 \vec{k} \text{ in 3 dimensions}),$$

where c_1, c_2, c_3 are arbitrary real constants. Indeed, each scalar component produces its own constant of integration, and grouping them component-wise gives a constant vector. This result is examinable in the sense that you must use it correctly: a GCE examiner will penalise a scalar constant added to a vector expression.

WARNING Scalar constant instead of vector constant

Two errors recur every year in candidates' scripts:

- Writing $\int \vec{v} dt = \vec{r}(t) + c$ with a scalar c . A scalar cannot be added to a vector: the constant must be a vector $\vec{c}$.
- Remembering the constant in one component only, e.g. writing $(2t^2 + c_1)\vec{i} + 3t\vec{j}$. Every component carries its own constant, so the second component must read $(3t + c_2)\vec{j}$.

A quick safety check: differentiate your answer. You must recover the integrand exactly, and the constants must disappear.